

Sensing Peripheral Devices with MC1489A Receiver

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Presented here is a circuit for monitoring the status of electrical or peripheral devices using the classical MC1489A receiver found in RS232 based devices. RS232 standard is still widely incorporated in embedded systems. Sometimes, there are unused or spare MC1489A receivers in RS232 interface circuits. Fortunately, these can be used for monitoring the status of electrical circuits or mechanical switches. This saves money and space and also adds some safety to embedded systems because receivers for RS232 have internal protection and Schmitt triggers

for better noise immunity. MC1489A captures information from peripheral devices that are not TTL/CMOS compatible but are in the working range of this receiver.

Circuit and working

Circuit diagram of the sensing peripheral devices with MC1489A receiver (IC1) is shown in Fig. 1. It consists of four examples for sensing the signals from peripheral devices such as simple DC motors (M1), incandescent lamps (BL1) and DC motor (M2) driven by a full-bridge circuit with four switches (S3-S6) to control bipolar DC voltages. Status of each device is indicated by four



PARTS LIST

Semiconductors:

- IC1 - MC1489A receiver
- D1-D5 - 1N4007 rectifier
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 2.2-kilo-ohm

Capacitors:

- C1-C3 - 0.1 μ F ceramic disk

Miscellaneous:

- S1-S6 - On/off toggle switch
- BL1 - 24V, 0.1A or similar bulb
- M1, M2 - 24V, 0.1A or similar DC motor

CON1, CON3,

- CON4 - 3-pin berg strip connector
- CON2 - 2-pin berg strip connector
- CON5 - 5-pin berg strip connector

LEDs (LED1 through LED4). For example, if switch S1 is closed, bulb BL1 will glow. Status of switching on the bulb is indicated by the glowing of LED2.

MC1489A converts bipolar signals into CMOS/TTL signals. It is a quad-line receiver with response controls (RC1 through RC4). Response control lines are not used here. However, these are included in this design to facilitate users for future applications or expansion.

LED1 through LED4 are connected

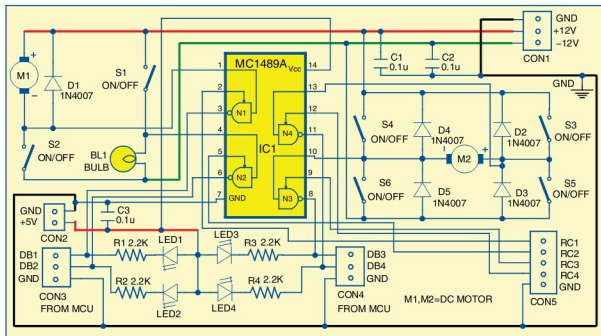


Fig. 1: Circuit diagram of the sensing peripheral devices with MC1489A receiver



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